

MH1812 Tutorial

Chapter 7: Set Theory

Q1: Deduce that the cardinality of the power set $P(S)$ of a finite set S with n element is 2^n .

Solution: The number of subsets of i elements from set S is $\binom{n}{i}$, hence the total number of subsets is $\sum_{i=0}^n \binom{n}{i} = (1+1)^n = 2^n$. $\square$

Q2: Consider the set $A = \{1, 2, 3\}$ and let $P(A)$ denote the power set of A .

- Compute the cardinality of $P(A)$ using the binomial theorem approach.
- Compute the cardinality of $P(A)$ using the counting approach.

Solution:

- $2^3 = 8$ (see Q1).
- The subsets are $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$, hence in total 8.

$\square$

Q3: Let $P(C)$ denote the power set of C . Given $A = \{1, 2\}$ and $B = \{2, 3\}$, determine:

$$P(A \cap B), P(A), P(A \cup B), P(A \times B).$$

Solution: $A \cap B = \{2\}$, hence $P(A \cap B) = P(\{2\}) = \{\emptyset, \{2\}\}$.

$P(A) = P(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

$A \cup B = \{1, 2, 3\}$, hence $P(A \cup B) = P(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.

$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$, hence

$P(A \times B) = \{$

$\emptyset,$

$\{(1, 2)\}, \{(1, 3)\}, \{(2, 2)\}, \{(2, 3)\},$

$\{(1, 2), (1, 3)\}, \{(1, 2), (2, 2)\}, \{(1, 2), (2, 3)\}, \{(1, 3), (2, 2)\}, \{(1, 3), (2, 3)\}, \{(2, 2), (2, 3)\},$

$\{(1, 2), (1, 3), (2, 2)\}, \{(1, 2), (1, 3), (2, 3)\}, \{(1, 2), (2, 2), (2, 3)\}, \{(1, 3), (2, 2), (2, 3)\},$

$\{(1, 2), (1, 3), (2, 2), (2, 3)\}\}$. $\square$

Q4: Prove by contradiction that for two sets A and B

$$(A - B) \cap (B - A) = \emptyset.$$

Solution: Prove by contradiction: assume $(A - B) \cap (B - A) \neq \emptyset$. Then there exists some element $x \in (A - B) \cap (B - A)$. This means x belongs to both $(A - B)$ and $(B - A)$. Then $x \in A$ (since $x \in A - B$), and $x \notin A$ (since $x \in B - A$), hence contradiction. $\square$

Q5: Let $P(C)$ denote the power set of C . Prove that for two sets A and B

$$P(A) = P(B) \iff A = B$$

Solution: To prove $P(A) = P(B) \iff A = B$, we need to prove both $P(A) = P(B) \Rightarrow A = B$ and $P(A) = P(B) \Leftarrow A = B$:

- When $P(A) = P(B)$, the subsets of single element should be the same for both, which means A and B are the same.
- When $A = B$, it implies $P(A) = P(B)$ due to the definition.

$\square$

Q6: Let $P(C)$ denote the power set of C . Prove that for two sets A and B

$$P(A) \subseteq P(B) \iff A \subseteq B.$$

Solution: To prove $P(A) \subseteq P(B) \iff A \subseteq B$, we need to prove both $P(A) \subseteq P(B) \rightarrow A \subseteq B$ and $P(A) \subseteq P(B) \Leftarrow A \subseteq B$:

- $P(A) \subseteq P(B)$, so the subsets of single element in $P(A)$ are also in $P(B)$, hence all elements in A are also in B , i.e., $A \subseteq B$.
- For any $X \in P(A)$, $X \subseteq A \subseteq B$, i.e., $X \subseteq B$, hence $X \in P(B)$.

$\square$

Q7: Show that the empty set is a subset of any set¹.

Solution: Recall the definition of subset: $Y \subseteq X$ means by definition that $\forall x, (x \in Y \rightarrow x \in X)$. Now take Y to be the empty set $\emptyset$. Since $x \in Y$ is necessarily false (one cannot take any x from the empty set), then the conditional statement is vacuously true. $\square$

Q8: Show that for two sets A and B

$$A \neq B \equiv \exists x[(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)].$$

¹Yes, the empty set is a subset of itself.

Solution:

$$\begin{aligned} A \neq B &= \neg \forall x (x \in A \leftrightarrow x \in B) \\ &\equiv \exists x \neg (x \in A \leftrightarrow x \in B) \text{ (negation of universal quantifier)} \\ &\equiv \exists x \neg [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)] \text{ (definition)} \\ &\equiv \exists x [(\neg(x \in A \rightarrow x \in B)) \vee (\neg(x \in B \rightarrow x \in A))] \text{ (DeMorgan)} \\ &\equiv \exists x [(\neg(x \notin A \vee x \in B)) \vee (\neg(x \notin B \vee x \in A))] \text{ (Conversion)} \\ &\equiv \exists x [(x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)] \text{ (DeMorgan)} \end{aligned}$$

□

Q9: Prove that for the sets A, B, C, D

$$(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D).$$

Does the equality hold?

Solution: For any $x \in (A \times B) \cup (C \times D)$, $x \in (A \times B)$ or $x \in (C \times D)$. When $x \in (A \times B)$, $x = (x_1, x_2) \in (A \times B)$, by definition $x_1 \in A \subseteq A \cup C$, and $x_2 \in B \subseteq (B \cup D)$, hence $x \in (A \cup C) \times (B \cup D)$. We can prove the same when $x \in (C \times D)$.

The equality does NOT necessarily hold: take $A = [-1, 0]$, $B = [-1, 0]$, $C = [0, 1]$, $D = [0, 1]$ (all the sets are intervals, that is $[a, b]$ means the interval from a to b). Then

$$[-1, 0] \times [-1, 0] \neq [-1, 1] \times [-1, 1].$$

□

Q10: Does the equality

$$(A_1 \cup A_2) \times (B_1 \cup B_2) = (A_1 \times B_1) \cup (A_2 \times B_2)$$

hold?

Solution: It does NOT hold. For example, take $A_1 = \{0\}$, $A_2 = \{1\}$, $B_1 = \{0\}$, $B_2 = \{1\}$, then

$$(A_1 \cup A_2) \times (B_1 \cup B_2) = \{0, 1\} \times \{0, 1\} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$

while

$$(A_1 \times B_1) \cup (A_2 \times B_2) = \{(0, 0), (1, 1)\}.$$

□

Q11: How many subsets of $\{1, \dots, n\}$ are there with an even number of elements? Justify your answer.

Solution: 2^{n-1} .

Let X, Y denote the number of subsets of even, or odd sizes respectively, we are to prove $X = Y$.

$$\begin{aligned} 0 &= (1 - 1)^n \\ &= \sum_{i=0}^n \binom{n}{i} (-1)^i \\ &= \sum_{i=0,2,4,\dots} \binom{n}{i} - \sum_{i=1,3,5,\dots} \binom{n}{i} \\ &= X - Y \end{aligned}$$

Hence, $X = Y$, and since $X + Y = 2^n$, $X = Y = 2^{n-1}$. □

Q12: Prove the following set equality:

$$\{7a + 9b \mid a, b \in \mathbb{Z}\} = \mathbb{Z}.$$

Solution: First of all, it is easy to see, all elements in the LHS are integers, hence $\text{LHS} \subseteq \mathbb{Z}$. Then we have to prove every integer is also in LHS. Note for $7 \cdot (-5) + 9 \cdot 4 = 1$, so any integer n can be expressed as $n \cdot (7 \cdot (-5) + 9 \cdot 4) = 7 \cdot (-5n) + 9 \cdot (4n)$, i.e., with $a = -5n$ and $b = 4n$. □

Q13: Let A, B, C be sets. Prove or disprove the following set equality:

$$A - (B \cup C) = (A - B) \cap (A - C).$$

Solution: $\text{LHS} = A - (B \cup C) = A \cap \overline{B \cup C} = A \cap \overline{B} \cap \overline{C}$
 $\text{RHS} = (A \cap \overline{B}) \cap (A \cap \overline{C}) = (A \cap A) \cap \overline{B} \cap \overline{C} = A \cap \overline{B} \cap \overline{C} = \text{LHS}.$ □

Q14: For all sets A, B, C , prove that

$$\overline{(A - B) - (B - C)} = \overline{A} \cup B.$$

using set identities.

Solution: $\text{LHS} = \overline{(A \cap \overline{B}) \cap \overline{B \cap C}} = \overline{(A \cap \overline{B}) \cap (\overline{B} \cup C)} = \overline{A \cap (\overline{B} \cap (\overline{B} \cup C))} = \overline{A \cap \overline{B}} = \overline{A} \cup B = \text{RHS}.$ □

Q15: This exercise is more difficult. For all sets A and B , prove $(A \cup B) \cap \overline{A \cap B} = (A - B) \cup (B - A)$ by showing that each side of the equation is a subset of the other.

Solution:

$$\begin{aligned}
x \in (A \cup B) \cap \overline{A \cap B} &\equiv x \in (A \cup B) \wedge x \in \overline{A \cap B} \\
&\equiv (x \in A \wedge x \notin A \cap B) \vee (x \in B \wedge x \notin A \cap B) \\
&\equiv (x \in A - B) \vee (x \in B - A) \\
&\equiv x \in (A - B) \cup (B - A)
\end{aligned}$$

□

Q16: The symmetric difference of A and B , denoted by $A \triangle B$, is the set containing those elements in either A or B , but not in both A and B .

1. Prove that $(A \triangle B) \triangle B = A$ by showing that each side of the equation is a subset of the other.
2. Prove that $(A \triangle B) \triangle B = A$ using a membership table.

Solution:

- First we show that $(A \triangle B) \triangle B \subseteq A$. Suppose that $x \in (A \triangle B) \triangle B$ and we shall show that $x \in A$. Since $x \in (A \triangle B) \triangle B$, it must hold that $x \in (A \triangle B) \setminus B$ or $x \in B \setminus (A \triangle B)$. We discuss the two cases.
 - $x \in (A \triangle B) \setminus B$: we know that $x \in A \triangle B$ and $x \notin B$. Since $x \in A \triangle B$, we know that $x \in A \setminus B$ or $x \in B \setminus A$. Since $x \notin B$, it cannot hold that $x \in B \setminus A$ and it must thus hold that $x \in A \setminus B$ and therefore $x \in A$.
 - $x \in B \setminus (A \triangle B)$: we know that $x \in B$ and $x \notin A \triangle B$. Note that $x \notin A \triangle B$ means that $x \in A \cap B$ or $x \notin A \cup B$. Since $x \in B$, it cannot hold that $x \notin A \cup B$ and it must thus hold that $x \in A \cap B$. Therefore, $x \in A$.

Hence we have $x \in A$ in either case. This shows that $(A \triangle B) \triangle B \subseteq A$.

Next we show that $A \subseteq (A \triangle B) \triangle B$. Let $x \in A$ and we shall show that $x \in (A \triangle B) \triangle B$. We discuss whether x belongs to B or not.

- $x \in B$: In this case $x \in A \cap B$ and thus $x \notin A \triangle B$, which implies that x is contained in exactly one of $A \triangle B$ and B . Hence $x \in (A \triangle B) \triangle B$.
- $x \notin B$: In this case $x \in A \triangle B$, and again x is contained in exactly one of $A \triangle B$ and B , that is, $x \in (A \triangle B) \triangle B$.

Hence we have $x \in (A \triangle B) \triangle B$ in either case. This shows that $A \subseteq (A \triangle B) \triangle B$.

- By a membership table:

A	B	$A \triangle B$	$(A \triangle B) \triangle B$
0	0	0	0
0	1	1	0
1	0	1	1
1	1	0	1

□

Q17: In a fruit feast among 200 students, 88 chose to eat durians, 73 ate mangoes, and 46 ate lychees. 34 of them had eaten both durians and mangoes, 16 had eaten durians and lychees, and 12 had eaten mangoes and lychees, while 5 had eaten all 3 fruits. Determine, how many of the 200 students ate none of the 3 fruits, and how many ate only mangoes?

Solution: Lets denote the set of students who ate durians, mangoes, lychees as D, M, L , respectively. From the question we have

$$|D| = 88, |M| = 73, |L| = 46, |D \cap M| = 34, |D \cap L| = 16, |M \cap L| = 12, |D \cap M \cap L| = 5.$$

Now, we want to find out $|\overline{D \cup M \cup L}|$.

$$\begin{aligned} |D \cup M \cup L| &= |D| + |M| + |L| - |D \cap M| - |D \cap L| - |M \cap L| + |D \cap M \cap L| \\ &= 88 + 73 + 46 - 34 - 16 - 12 + 5 \\ &= 150, \end{aligned}$$

$$\text{hence } |\overline{D \cup M \cup L}| = 200 - 150 = 50.$$

The set of students who ate only mangoes can be expressed as $M \cap \overline{D} \cap \overline{L}$, i.e., those who ate mangoes, but not durians nor lychees.

$$\begin{aligned} |M \cap \overline{D} \cap \overline{L}| &= |M| - |M \cap D| - |M \cap L| + |M \cap D \cap L| \\ &= 73 - 34 - 12 + 5 \\ &= 32. \end{aligned}$$

□

Q18: Let A, B, C be sets. Prove or disprove the following set equality:

$$A \times (B - C) = (A \times B) - (A \times C).$$

Solution: For any $x = (x_1, x_2) \in A \times (B - C)$, by definition it means $x_1 \in A$ and $x_2 \in (B - C)$ and $x_2 \in (B - C) \iff x_2 \in B \wedge x_2 \notin C \iff (x_1, x_2) \in A \times B \wedge (x_1, x_2) \notin A \times C \iff (x_1, x_2) \in (A \times B) - (A \times C)$. □